

Qatar-1b

R-0023

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260522 · created 2026-07-08T10:14:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461182.8827 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.054
Transit depth (Rp/Rs)^2	3.01 +/- 1.88 [%]
Orbital Inclination (inc)	83.18 +/- 2.87
Airmass coefficient 1 (a1)	3.47 +/- 1.91
Airmass coefficient 2 (a2)	-0.94 +/- 0.95
Scatter in the residuals of the lightcurve fit is	54.82 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	3.0
Transit Duration (day)	0.059 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.054 vs 0.1463 (deviation 0.49σ)
Duration fit vs published	0.059 d vs 0.0692 d (deviation 0.41σ)
Mid-time O–C	-8.1 min
Depth SNR	3.1
Residual scatter	54.82 %

Light curve

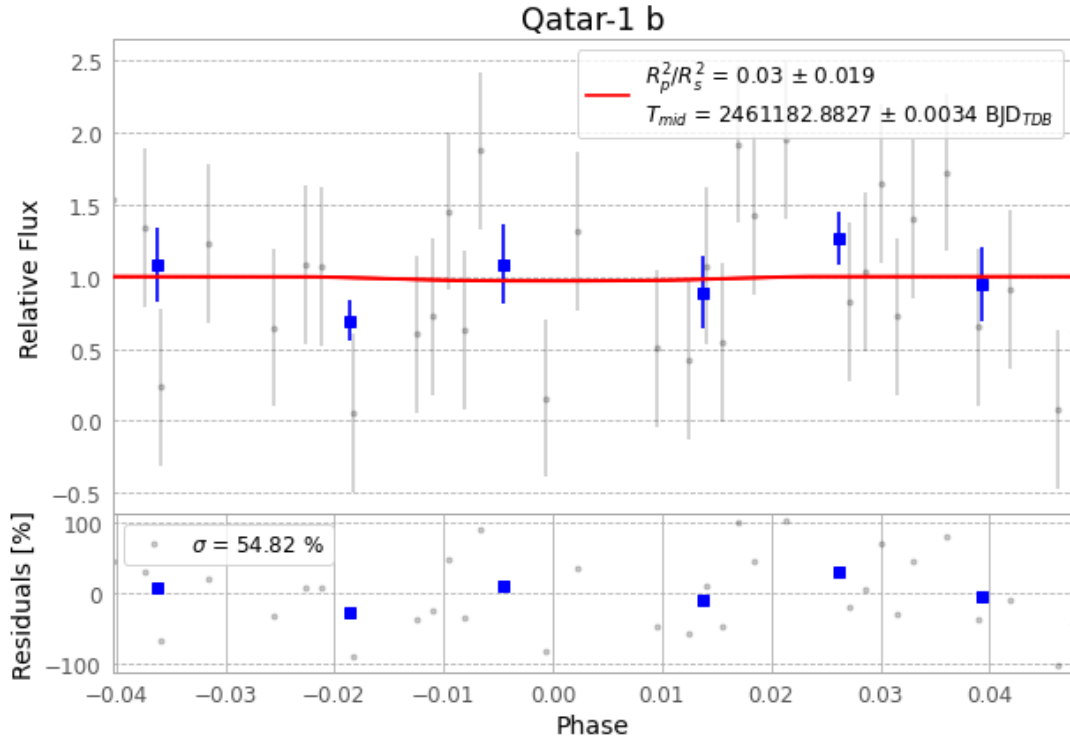


Figure 1 — FinalLightCurve_Qatar-1 b_2026-05-22.png (SHA-256 e7087401718a9d24...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c3a9d3b929c2cdf1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

63 FITS frames (41 MB), Qatar-1260522074815.FITS → Qatar-1260522105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260522.log (28738e39b053...), FinalLightCurve_Qatar-1 b_2026-05-22.png (e7087401718a...), FinalLightCurve_Qatar-1 b_2026-05-22.csv (5308632d96a3...)

Compute resources

Wall time 122.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.